UNIT 2 / DAY 2 • EMERGENCY PAPER VERSION

Industrial Transformation

Username: _____ Class: _____ Date: _____

Username rule: at least two letters followed by at least two numbers (RIVER27). Do not use your real name or school student ID. Keep the same username when returning to the HTML.

Use this packet if the student HTML is unavailable. It contains the pre-Boost lesson and space for your responses. Read the gallery evidence on the wall; it is not reproduced here. The interview and formative reflection are in a separate document.

Paper completion: _____ / 39 response fields + _____ / 8 vocabulary matches. Teacher vocabulary check: _____ / 8.

01 / THE BIG INQUIRY · 0-12 MIN

More possibilities. For whom?

Industrial Transformation

How did industrial technology expand what people could do? What benefits and harms did it help create, who experienced them, and how did people respond?

I can explain what the Industrial Revolution was, how it expanded human possibilities, and investigate what determined who benefited.

What changed?

The Industrial Revolution was a long transformation in how people produced goods: machines, new energy sources, and factories reorganized work. It began in Britain in the eighteenth century and spread unevenly. In this lesson, we focus on the United States, especially 1865-1920.

Machines increased what workers could produce. Steam engines made coal a source of mechanical power. Factory owners organized workers and machines in one production system. These changes expanded capability and created new conflicts over work, wealth, and the environment.

Quick partner check

One partner explains how a machine changes production; the other explains why the way work is organized matters. Then switch.

WATCH WITH YOUR CLASS · ABOUT 5 MIN

How inventions change history

The cotton gin is an earlier example, patented in 1794 after its development in 1793. It separated seeds from cotton; it did not pick cotton. Greater processing capacity increased demand for cotton and helped expand slavery. Watch for the connection between an invention and the institutions around it.

Open the TED-Ed video https://www.youtube.com/watch?v=0SMNYivhGsc

Video: Kenneth C. Davis / TED-Ed. Your teacher will play it.

What became easier with the cotton gin? Why did that fail to make life better for everyone?

Name the capability and explain who controlled how it was used.

Explain industrialization in your own words. Include machines, energy, and the organization of work.

02 / LOOK AT THE SCALE · 12-17 MIN

When are we studying?

Our focus is the United States from 1865 to 1920. The shaded area shows where those 55 years fit within the past 2,000 years. Equal horizontal distances represent equal amounts of time.

1 CE–2000: proportional time



Shaded interval: United States, 1865–1920

Use the buttons to zoom in on our study period. The graphs below show changes in production, survival, and lighting; this timeline shows when we are studying.

Predict, then discuss

Which would change your daily possibilities most: more goods, better chances of survival, or cheaper light? Choose one and defend it to your partner. You can revise your thinking after the evidence.

03 / THREE ANCHOR INVESTIGATIONS

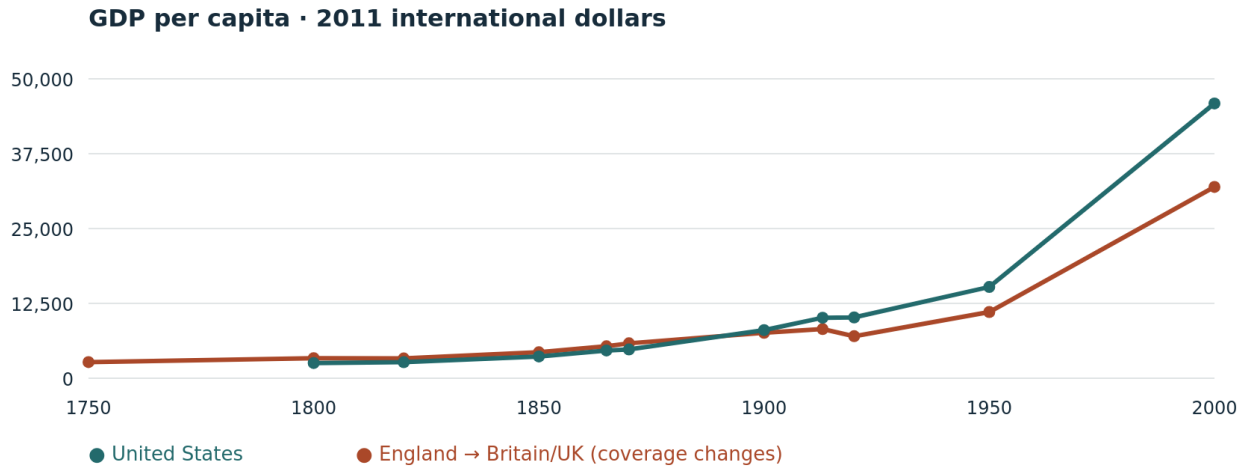
What became possible?

For each investigation: identify a pattern, use a number, and ask what the measure leaves out. A trend shows a change; it does not by itself prove its cause.

PRODUCTION

More output per person

GDP per capita divides the value of economic production by population. It is an average, not a worker's pay. The values here use purchasing power adjustments and 2011 international dollars.



Selected historical estimates, Maddison Project Database 2023. The early comparison is England (through 1700), then Britain/UK; coverage changes. Straight segments connect benchmarks, not observations for every intervening year. These are not world averages.

Read the benchmark values

Year	England / Britain / UK*	United States
1270	1,192	Not shown
1500	1,697	Not shown
1600	1,691	Not shown
1700	2,412	Not shown
1750	2,702	Not shown
1800	3,343	2,545
1820	3,306	2,674
1850	4,332	3,632
1865	5,333	4,637
1870	5,829	4,803
1900	7,594	8,038
1913	8,212	10,108
1920	7,017	10,153
1950	11,061	15,240
2000	31,946	45,886

*Early geography changes. Display rounded; calculations use source precision.

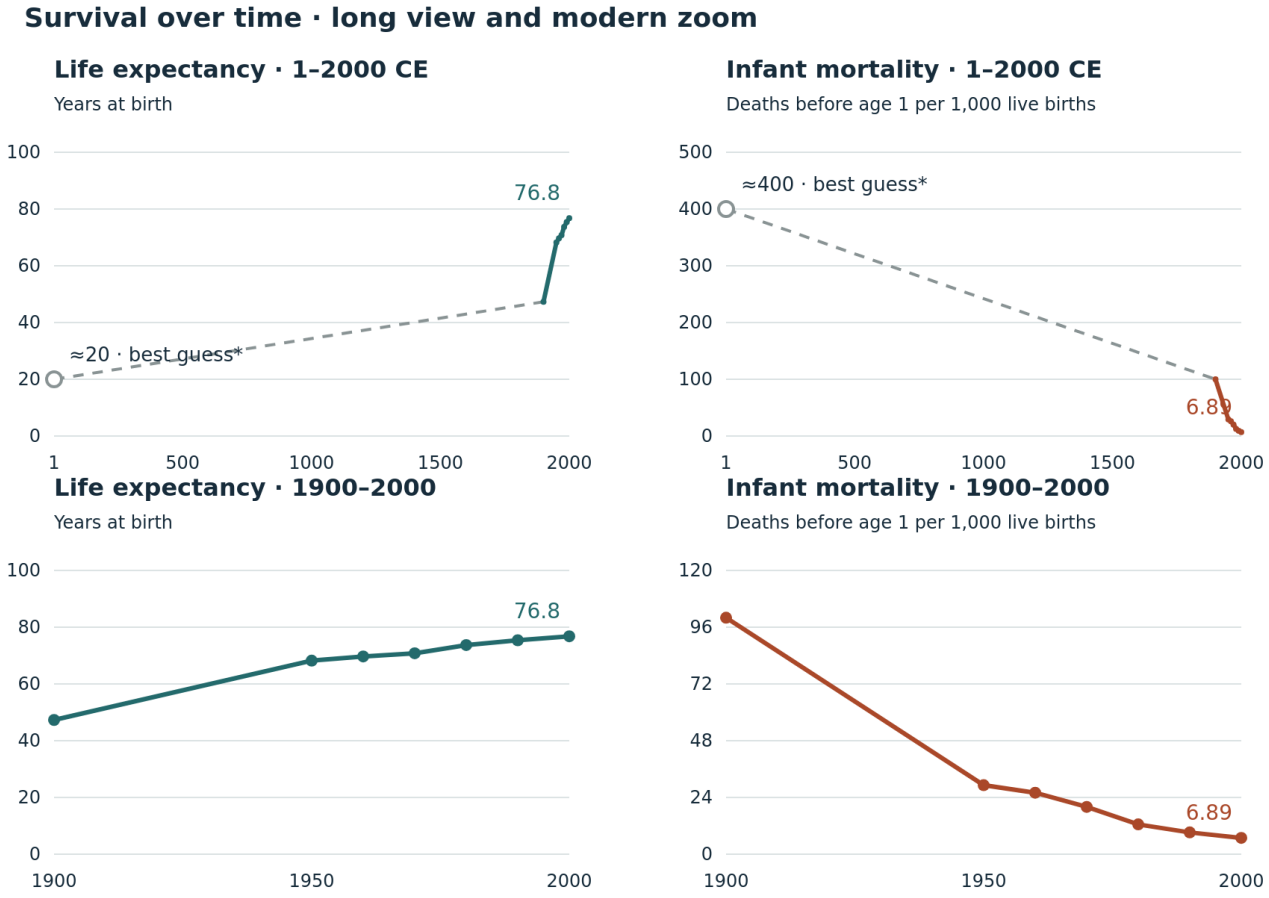
Use two U.S. values to describe the change in production per person from 1865 to 1920. Why can't this average tell us who benefited?

The change is about 119%. Explain what an average can hide.

SURVIVAL

More chances to live

Life expectancy at birth summarizes survival under a year's death rates. High infant mortality pulled the average down; a value of 47 years does not mean most adults died at 47.



*Ancient reference: Roman-era reconstruction placed at year 1 for comparison.

Dashed bridge = illustrative connection, not observed centuries. Solid lines = U.S. benchmarks.

Year 1: best guesses, not measured statistics. We use about 20 years of life expectancy and about 400 infant deaths per 1,000 births as an illustrative ancient reference from one Roman-era reconstruction (Langner, 1998). Its evidence comes from later Roman centuries; other reconstructions differ. These are not precise year-1 or world averages. The dashed bridge connects different populations for comparison; it does not show what happened in every intervening century. Solid lines show selected U.S. benchmarks: life expectancy 47.3 in 1900 to 76.8 in 2000; infant deaths about 100 to 6.89 per 1,000 births. Early registration coverage and estimation differ from later national statistics. The 1950-1990 infant rates use mortality/natality files; the 2000 value uses linked birth-infant death records. Sources: Langner; CDC/NCHS.

Imagine 1,000 births. The early estimate means about 100 infant deaths; the later rate means about 7. This comparison describes populations, not the fate of any particular child. Notice the recent rise in life expectancy and the downward bend in infant mortality. Sanitation, nutrition, public health, and medicine all helped; industrial technology alone does not explain the changes.

Read the data, then make an inference

Answer B1-B5. Short answers are enough; use the graph values when requested.

B1. Read the U.S. life-expectancy line. What are the values for 1900 and 2000? How many years did the estimate increase?

Subtract the earlier value from the later value. Include units.

B2. Out of 1,000 births, about how many infants died in 1900 and in 2000? About how many fewer infant deaths is that?

Round the 2000 rate to a whole infant for this comparison.

B3. Use one survival number to explain a new possibility for a family or community. Explain the connection; do not just repeat the number.

B4. A student says, “Life expectancy was 47.3 in 1900, so most adults died at about 47.” Explain what is wrong with that conclusion.

Consider how deaths in infancy affect the average.

B5. Why should we be more cautious about the year-1 point than the modern U.S. points? Give one reason the graph alone cannot prove factories caused better survival.

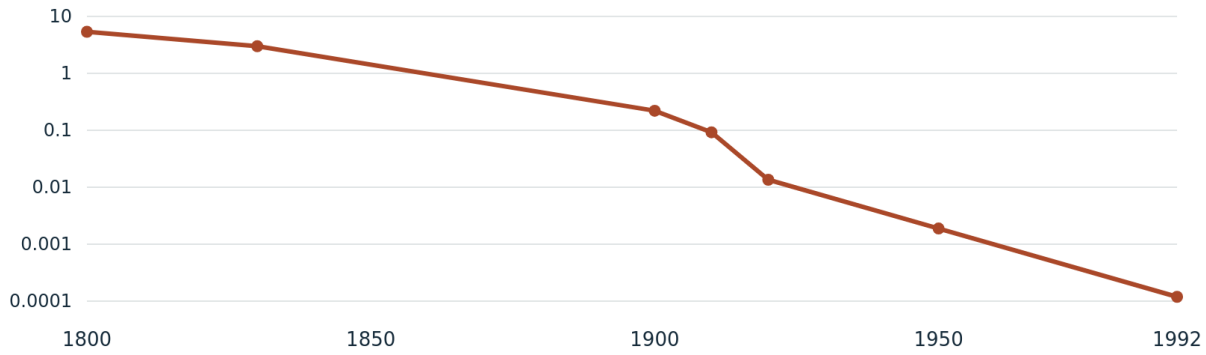
Read the source note: certainty, population, and possible causes.

EVERYDAY CAPABILITY

The cost of keeping the light on

Compare the labor time needed to pay the operating cost of the same light service: 1,000 lumen-hours. Think of a light producing 1,000 lumens for one hour.

Labor hours for 1,000 lumen-hours · logarithmic scale



Each equal vertical step represents a tenfold change; use the table for exact comparisons.

Year	Technology	Labor time
1800	Tallow candle	5.4 hours
1830	Tallow candle	3 hours
1900	Carbon-filament lamp	13.2 minutes
1910	Carbon-filament lamp	5.5 minutes
1920	Filament lamp	48.7 seconds
1950	Filament lamp	6.8 seconds
1992	Compact fluorescent lamp	0.43 seconds

Nordhaus, Table 1.6, pp. 52-53. Selected U.S. technologies, fuel/energy operating cost only; equipment and installation excluded. Wage measures change across dates. The final observation is 1992, not 2000. Lower cost does not demonstrate that every household had access.

Read the data, then make an inference

Answer C1-C5. Use the table for exact values; the graph uses a logarithmic scale.

C1. How much labor time paid for the same light service in 1800 and in 1900? Give both values and their units.

C2. Compare 1910 and 1920. Convert 5.5 minutes to seconds, then estimate how many times as much labor time the 1910 light required.

Divide the 1910 time in seconds by the 1920 time. Round to the nearest whole number.

C3. Imagine a student who works during the day. How might cheaper light change their opportunities? Use one number from the table to support your inference.

C4. A student says, “Cheap light meant every home had electric lighting.” What TWO additional things would a household need? Explain why this cost measure cannot establish universal access.

The table excludes equipment and installation.

C5. Describe one possible benefit and one possible harm of extending work or activity after dark. Who might experience each?

These are reasoned possibilities, not outcomes proved by this table.

Beyond the three anchors

Open all eight evidence cards. In each one, answer its short question using a specific detail. Then choose one card for the final explanation. Most card answers need only one sentence.

Literacy

20% to 10.7% illiterate

United States, 1870 to 1900, ages 14 and older. These historical categories describe basic reading/writing, not modern functional literacy. In 1900 the reported rates were 6.2% for White people and 44.5% for the source's combined “Black and other” category. Race and class are different dimensions; unequal educational opportunity matters.

NCES, 120 Years of Literacy.

What happened to the overall illiteracy rate from 1870 to 1900? Use the 1900 group figures to explain why the overall rate can hide unequal access to education.

One or two sentences. Use a detail from this card.

Schooling

51% to 75% enrolled

United States, ages 5-19, 1900 to 1940. Enrollment is not the same as attendance, school quality, or graduation. Notice that the second date extends beyond our 1865-1920 focus.

NCES, 120 Years of Literacy.

How many percentage points did enrollment rise? Why does being enrolled not prove that a student attended regularly or graduated?

One or two sentences. Use a detail from this card.

Poverty

A series beginning in 1959

The U.S. Census official poverty series begins in 1959. It cannot supply a comparable poverty rate for 1900. Its income thresholds measure one kind of deprivation; they do not capture every aspect of hardship. Later poverty measures are an extension of our inquiry, not direct evidence for the Gilded Age.

U.S. Census, historical poverty tables.

Why can't this official poverty series tell us the U.S. poverty rate in 1900? State its starting year.

One or two sentences. Use a detail from this card.

Energy per person

About 102 to 126 million Btu

United States, 1850 to 1900. Calculation: EIA estimated total primary energy divided by Census population. Totals were 2.357 and 9.587 quadrillion Btu; populations were 23,191,876 and 76,212,168. These estimates exclude direct mechanical water power and do not measure a household's energy bill or useful output.

EIA Table E1; U.S. Census 1850 and 1900.

How did estimated energy per person change from 1850 to 1900? Give one reason this does not tell us how much energy every household could use.

One or two sentences. Use a detail from this card.

Travel

Coast-to-coast in about a week

The first transcontinental railroad connection was completed in 1869. The New York-San Francisco journey could take about seven days, compared with months on an overland wagon journey. Route, fare, safety, and access mattered; this is not a universal travel-time average.

ASCE, Union Pacific Railroad historical landmark.

Compare the journey times. Name one opportunity faster rail travel might create and one factor that could keep someone from using it.

One or two sentences. Use a detail from this card.

Information delivery

A wire crossed the continent in 1861

The transcontinental telegraph linked the eastern and western United States in 1861. Messages no longer had to travel physically the entire distance. Rapid transmission did not guarantee immediate delivery to every home, and service had a cost.

National Park Service, Pony Express / telegraph history.

What changed about sending a message in 1861? Why might fast transmission still fail to produce immediate delivery to a home?

One or two sentences. Use a detail from this card.

Paid work hours

55-hour week / 49 paid hours

U.S. manufacturing in 1915: an average workweek of about 55 hours, compared with about 49 paid hours for production workers. These measures differ. Neither includes all farm work, unpaid household labor, or everyone's leisure time. We cannot subtract these numbers from 168 and call the remainder “free time.”

BLS, The life of American workers in 1915.

What does the 55-hour manufacturing workweek tell us? Why can't we use either hours figure to calculate everyone's leisure time?

One or two sentences. Use a detail from this card.

Household infrastructure

Available did not mean universal

The Vanderbilt mansion, occupied by 1899, had electricity, hot and cold running water, and flush toilets. This documented wealthy household shows what was possible, not what every home had. Infrastructure required equipment, local systems, and money.

National Park Service, Vanderbilt Mansion audio tour.

Name two conveniences in the Vanderbilt home. Explain the difference between a technology existing and everyone having access to it.

One or two sentences. Use a detail from this card.

Choose ONE supporting card. Name it, explain one change or problem it helps us understand, and state ONE thing its evidence cannot tell us.

Example structure: The ___ card shows ___. It cannot tell us ___ because ___.

04 / WHO BENEFITS?

A bigger economy can still leave a wider gap.

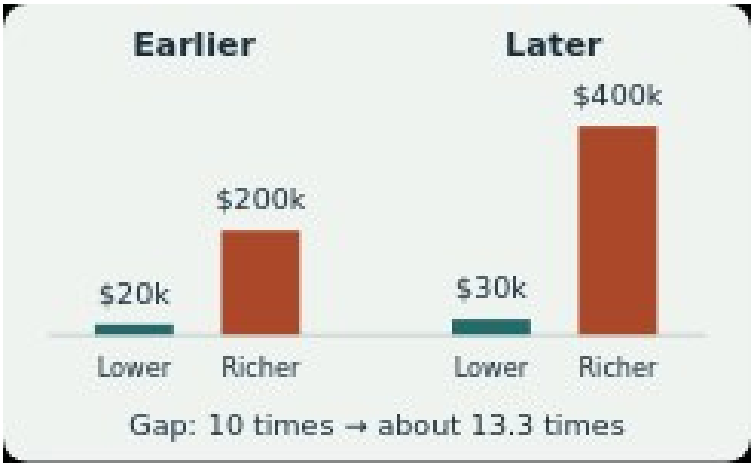
Absolute well-being



What can a person afford or do?

Food, lighting, travel, and health may improve even if the gap with richer people grows.

Relative position



How does income compare with others? Illustration: k means thousand dollars.

A larger gap can exist alongside improvements in both groups' lives. See the made-up example below.

Security and participation



Can people keep opportunities and have a say?

Can a person keep their job, obtain care, attend school, and influence decisions? More goods do not automatically create these protections.

Try the logic-with dollars

This is a made-up example, not historical wages. Compare annual household income. Assume prices, household size, and needs stay the same.

Household	Earlier income	Later income	Increase
Lower income	$20,000	$30,000	$10,000 = 50%
Richer	$200,000	$400,000	$200,000 = 100%

Absolute change: the change in a household's own income or purchasing power. A 50% increase means adding half of the starting amount: half of $20,000 is $10,000.

Relative position: how one household's income compares with another's. To compare, divide the richer household's income by the lower-income household's income in each year.

D1. Use the dollar table to explain BOTH arguments: “This clearly shows progress” and “This shows increasing unfairness.” Use a number for each argument.

Explain the two views fairly; you do not have to agree with both.

D2. How many times as much did the richer household earn earlier? How many times as much later? What happened to the relative gap?

Divide richer income by lower income. Round the later ratio to one decimal place.

Historical global estimates put the average income of the top 10% at about 18 times that of the bottom 50% in 1820, and about 41 times in 1910. These are global reconstructions, not a U.S. series. A ratio alone cannot tell us whether the bottom half's absolute income rose or fell. Source: World Inequality Report 2022, Figure 2.2.

Turn percentages and ratios into people and dollars

Percentage of people: in a group of 100, the top 10% means the 10 people with the highest incomes; the bottom 50% means the 50 with the lowest incomes. These labels do not say what percentage of all income those people receive.

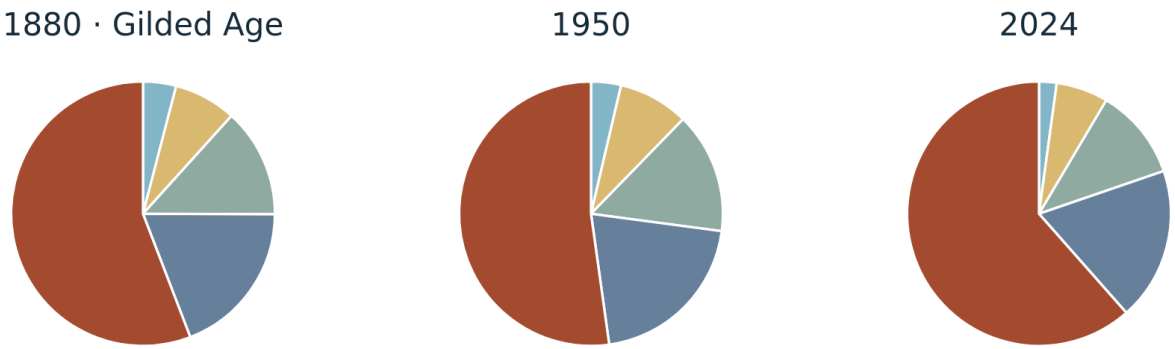
Income share: in a made-up community with $1,000,000 of total annual income, a group receiving 60% receives $600,000 altogether; a group receiving 7% receives $70,000 altogether. These are group totals, not each person's income.

Average-income ratio: if one group's average annual income were $10,000 and another's were $410,000, the second would be 41 times the first. These dollar amounts illustrate the math; they are not estimates of 1910 incomes.

Who received the income? Three U.S. snapshots

A quintile is one fifth of the population: 20 out of every 100 adults, ranked from lowest to highest income. Each pie shows how the total income was divided, not how the population was divided. With equal incomes, each quintile would receive 20%.

Income is a flow over a period, such as earnings and investment income during a year. Wealth is the value of assets minus debts at a point in time. These charts show income shares, not wealth or average dollar incomes.



Quintiles: lowest 20% → highest 20%, clockwise from the top. Exact shares in the table.

Income-ranked group	1880	1950	2024
Lowest 20%	4.06%	3.68%	2.16%
Second 20%	7.66%	8.63%	6.38%
Middle 20%	13.34%	14.80%	11.22%
Fourth 20%	19.10%	20.68%	18.72%
Highest 20%	55.84%	52.21%	61.52%
Top 10% (inside highest 20%)	41.00%	38.33%	46.76%
Top 1% (inside top 10%)	14.50%	16.39%	20.73%

Comparable concept throughout: estimated U.S. pretax national income shares among adults age 20+, using WID's equal-split-adult method (income shared equally within a couple). This is broader than wages or Census household money income; it includes labor and capital income and pension/unemployment insurance benefits, before income taxes and most redistribution. The same adult can belong to the top 20%, top 10%, and top 1%: do not add those rows together.

Source: World Inequality Database, U.S. bulk data retrieved October 21, 2026; sptincj992 (the bulk file's notation for sptinc992j). Quintiles calculated by subtracting cumulative shares. 1880 is a low-confidence, model-based Gilded Age reconstruction, not a modern survey; 1950 and 2024 are also estimates. 2024 is the latest year in this comparison, not a claim about 2026. Methods and source quality differ across periods; these are snapshots, not an annual trend. See WID's historical methods. Other researchers' definitions and estimates may differ.

https://wid.world/data/

D3. Compare the highest-income 20% in 1880, 1950, and 2024. Which year has the largest share and which has the smallest? Use the percentages. Then describe the change for the lowest-income 20%.

D4. In 2024, what share went to the top 1%? What share went to the top 10%? Why would adding those two shares count some income twice?

D5. If a group's share of income falls, does that prove its members have fewer dollars or worse living conditions? Explain what other information you would need.

Discuss: what would make a distribution fair?

Should fairness mean equal incomes, rewards for contributions, equal opportunity, or making sure everyone has enough? Consider more than one standard. A pie chart cannot choose a moral standard for us.

D6. Choose a fairness standard. Using one chart value, explain whether the distribution raises a concern under that standard. Give a reasonable counterargument or a fact you would need before judging.

There is no required political position. Support your reasoning and consider another view.

Compare possibilities, then ask about access.

Three housing examples-not a ranking of people's worth, happiness, or communities. The photos show different places and dates and are not to the same physical scale. An exterior cannot tell us a family's income, utilities, or daily experience.

What do we mean by poverty?

Absolute poverty: having resources below a defined minimum standard of basic needs, such as food, clothing, and shelter. For comparisons over time, an absolute line is held constant in purchasing power, with adjustments for prices. People still have to decide which needs and minimums count.

Relative poverty: having income far below the typical income in one's society. One common measure is below 50% of the median, after accounting for household size. The median is the middle income when incomes are ranked. A relative threshold can rise as society becomes richer.

These definitions answer different questions. Having electricity or a phone does not by itself establish whether someone is above either poverty line.

Definitions: OECD, Poverty rate; World Bank, absolute and relative poverty. The numbers below are fictional teaching examples, not official U.S. thresholds.

https://www.oecd.org/en/data/indicators/poverty-rate.html

DOCUMENTED HOME / OCCUPIED BY 1899

A wealthy household



Frederick and Louise Vanderbilt's mansion, Hyde Park, New York. Modern photograph: National Park Service (NPS photo).

The Hyde Park mansion had electricity, hot and cold running water, and flush toilets. These conveniences existed, but this exceptional household cannot represent most Americans.

Historical conveniences: NPS audio-tour transcript.

https://www.nps.gov/vama/learn/historyculture/audio-tour-of-the-mansion.htm

A mill-working household



Homes of cotton-mill workers, Lydia Mills, Clinton, South Carolina, December 2, 1908. Lewis W. Hine / Library of Congress, nclc.01477; no known publication restrictions.

This documented mill-community example helps us imagine a teenager's home. It is not our fictional interview character's actual home in Vermont, or a picture of every mill family. Choices depended on wages, work demands, school access, and local services.

A household with a limited income



Poe Homes, Baltimore, March 6, 2018. Eli Pousson / Baltimore Heritage; photograph, CC0. HABC identifies it as public housing.

Apartments are homes for people across income levels. This public-housing example illustrates one option; lower-income households also live in many other kinds of homes.

For our fictional comparison, imagine a household with electricity, school access, and an internet-connected phone, but rent pressure or insecure work. The photo does not establish these facts about its residents.

One household, two definitions

Imagine a community of same-sized households. One household has $26,000 of annual after-tax income. A made-up basic-needs poverty line is $24,000; the community median is $60,000. Use 50% of the median as the relative line. Assume the basic-needs line accurately reflects local prices for this exercise.

D7. Is the $26,000 household below the absolute line? Calculate the relative line: is the household below that one? Explain how both answers can be true.

D8. Suppose the household's income and prices stay the same, but the community median rises to $70,000. What happens to its absolute position and the relative poverty line? Does its buying power change?

D9. Should poverty mean lacking basic necessities, lacking what is needed to participate in society, or both? Defend your definition, address another view, and explain why the housing photos alone cannot decide who is poor.

Use a fictional example or lesson evidence. Do not disclose your own household finances.

05 / GALLERY WALK

Find the mechanisms.

The charts show outcomes. Now investigate evidence on the wall to explain how change happened and who experienced its benefits and costs.

Start at the station your teacher assigns. Move when directed; visit all six.

At each station, read the wall card and examine its evidence. Record one specific observation and a question or possible implication.

Return to your seat. Choose two stations that help explain one outcome from the charts.

Tell your partner how the evidence supports your explanation. Your partner asks, “What evidence would make you reconsider?” Revise or qualify your claim.

Wall-card evidence is available at the stations. These fields are your observation record.

Station 1

Station 1: record one specific observation from the wall evidence and one question or possible implication.

Station 2

Station 2: record one specific observation from the wall evidence and one question or possible implication.

Station 3

Station 3: record one specific observation from the wall evidence and one question or possible implication.

Station 4

Station 4: record one specific observation from the wall evidence and one question or possible implication.

Station 5

Station 5: record one specific observation from the wall evidence and one question or possible implication.

Station 6

Station 6: record one specific observation from the wall evidence and one question or possible implication.

Choose TWO station numbers. Explain how their evidence helps account for ONE outcome from the charts. Identify a benefit and a cost or unequal effect.

Station __ suggests __. Combined with station __, this could explain __ because __.

What did your partner challenge? Revise your explanation or state what additional evidence you would need.

06 / RETRIEVAL CHECK

Match the language to the idea.

Try from memory first. Match each term to a definition. After checking, revisit terms you missed and try again. The check records your first score and your latest score.

Definition bank

A. The shared systems and facilities that support daily life.

B. An estimate of average years of life under a population's death rates.

C. The process through which a cause can produce an effect.

D. A shift toward machine production, new energy sources, and reorganized work.

E. Unequal distribution of resources or opportunities among people.

F. Deaths before age one, usually measured per 1,000 live births.

G. The value of economic production divided by population.

H. Output produced for a given amount of input, such as labor time.

Term	Definition letter
Industrialization	_____

Term	Definition letter
GDP per capita	_____
Life expectancy	_____
Infant mortality	_____
Productivity	_____
Infrastructure	_____
Inequality	_____
Mechanism	_____

Have your teacher check your matches. Revisit the lesson and revise any you missed.

EVIDENCE NOTES

Sources and boundaries

Historical measures are reconstructions or estimates. Use the labels and notes, especially when comparing different populations or dates.

WID, comparable U.S. income shares: 1880, 1950, 2024 - Pretax national income, equal-split adults age 20+; bulk series sptincj992. Quintiles derived from cumulative shares. Historical reconstruction and modern estimates; income, not wealth.

https://wid.world/data/

WID, historical distribution methods (2026 revision) - Documents benchmark reconstruction and historical U.S. top shares.

https://wid.world/document/wid-income-and-wealth-distributional-series-updated-and-extended-coverage-1800-2024-world-inequality-lab-technical-note-2025-10/

OECD, Poverty rate - Relative line: half median household income; compare household sizes appropriately.

https://www.oecd.org/en/data/indicators/poverty-rate.html

World Bank, absolute and relative poverty concepts - Absolute lines fixed in purchasing power; relative lines tied to living standards. Dollar scenarios here are fictional, not official thresholds.

https://blogs.worldbank.org/en/opendata/updating-world-banks-societal-poverty-line-2017-purchasing-power-parities

NPS, Vanderbilt Mansion photograph - Frederick and Louise Vanderbilt's Hyde Park home; modern photograph credited NPS.

https://www.nps.gov/places/vanderbilt-mansion.htm

Lewis W. Hine / Library of Congress, Lydia Mills housing - Clinton, South Carolina, December 2, 1908; nclc.01477. No known publication restrictions. A mill-community example, not the Vermont character's actual home.

https://www.loc.gov/item/2018674015/

Eli Pousson / Baltimore Heritage, Poe Homes photograph - March 6, 2018; CC0. The photograph does not establish household income, utilities, or happiness.

https://commons.wikimedia.org/wiki/File:Apartments,_Poe_Homes_(1940),_800_W._Lexington_Street,_Baltimore,_MD_21201_(38883457860).jpg

Housing Authority of Baltimore City, Transform Poe - Confirms Poe Homes was a public-housing community. It is one housing example, not representative of all limited-income families.

https://www.habc.org/habc-information/programs-departments/planning-development/transform-poe/

G. Langner (1998), Roman-era survival reconstruction - One contested model: about 20 years at birth and nearly 400 infant deaths per 1,000. Used as an illustrative year-1 best guess; evidence from later Roman centuries, not a measured year-1/global statistic.

https://pubmed.ncbi.nlm.nih.gov/12178163/

CDC/NCHS, Health United States 2019, Table 4 - Selected life expectancy benchmarks, 1900-2000.

https://www.cdc.gov/nchs/data/hus/2019/004-508.pdf

CDC/NCHS, Health United States 2016, Table 11 - Selected infant mortality benchmarks, 1950-1990: 29.2, 26.0, 20.0, 12.6, 9.2 per 1,000 live births.

https://www.cdc.gov/nchs/data/hus/2016/011.pdf

Maddison Project Database 2023 - GDPpc sheet, GBR and USA. 2011 international dollars. Early GBR coverage changes; selected benchmarks only. Underlying credits: Broadberry et al. (2015), British Economic Growth 1270-1870; Feinstein (1972), National Income Expenditure and Output of the United Kingdom 1855-1965; Sutch (2006), National Income and Product; Prados de la Escosura (2009), Lost Decades?; Maddison (1995), Monitoring the World Economy. Source and original-source sheets give coverage. MPD: Bolt and van Zanden (2024), DOI 10.1111/joes.12618.

https://www.rug.nl/ggdc/historicaldevelopment/maddison/releases/maddison-project-database-2023?lang=en

William D. Nordhaus, Do Real-Output and Real-Wage Measures Capture Reality? The History of Lighting Suggests Not - Table 1.6, printed pages 52-53. Labor hours per 1,000 lumen-hours; selected technologies and changing wage measures.

https://www.nber.org/system/files/chapters/c6064/c6064.pdf

CDC/NCHS, life expectancy comparison - 47.3 (1900 registration states) and 76.8 (2000 national).

https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6019a5.htm

CDC/NCHS, infant mortality - Approximate early rate and 6.89 in 2000; infant deaths per 1,000 live births.

https://www.cdc.gov/nchs/products/databriefs/db09.htm

NCES, 120 Years of Literacy - Illiteracy: ages 14+, 20.0% (1870), 10.7% (1900); racial column is “Black and other,” not Black alone. Enrollment: ages 5-19, 51% (1900), 75% (1940).

https://nces.ed.gov/naal/lit_history.asp

NCES, 120 Years of American Education - Statistical portrait underlying the historical education summary.

https://nces.ed.gov/pubs93/93442.pdf

U.S. Census, historical poverty tables - Official series begins in 1959; no comparable 1900 rate shown.

https://www.census.gov/data/tables/time-series/demo/income-poverty/historical-poverty-people.html

EIA, estimated primary energy, Table E1 - 1850 and 1900 totals divided by Census population; direct mechanical water power excluded.

https://www.eia.gov/totalenergy/data/annual/txt/ptb1601.html

U.S. Census, 1850 population - 23,191,876.

https://www.census.gov/programs-surveys/decennial-census/decade.1850.html

U.S. Census, 1900 population - 76,212,168.

https://www.census.gov/programs-surveys/decennial-census/decade.1900.html

ASCE, Union Pacific Railroad - Historic landmark summary; approximately seven-day transcontinental travel.

https://www.asce.org/about-civil-engineering/history-and-heritage/historic-landmarks/union-pacific-railroad/

NPS, Pony Express history - Transcontinental telegraph completed in 1861.

https://www.nps.gov/poex/learn/historyculture/index.htm

BLS, The life of American workers in 1915 - Manufacturing workweek hours and paid production-worker hours are different measures.

https://www.bls.gov/opub/mlr/2016/article/the-life-of-american-workers-in-1915.htm

NPS, Vanderbilt Mansion audio tour - Documented household conveniences, 1899; not representative of all households.

https://www.nps.gov/vama/learn/historyculture/audio-tour-of-the-mansion.htm

World Inequality Report 2022, Chapter 2 - Figure 2.2: global average-income ratio 18 (1820) and 41 (1910). Population-group percentages differ from income shares. Dollar examples in this lesson are hypothetical.

https://wir2022.wid.world/chapter-2/

TED-Ed, How inventions change history (for better and for worse) - Opening cotton-gin example; publisher description verified.

https://ed.ted.com/lessons/how-inventions-change-history-for-better-and-for-worse-kenneth-c-davis

National Archives, cotton gin patent - Patent issued 1794; invention developed in 1793.

https://www.archives.gov/education/lessons/cotton-gin-patent

Vocabulary support: use this in place of hover definitions

absolute: Described on its own scale, rather than as a position compared with others.

absolute poverty: Resources below a defined minimum basic-needs standard, held constant in purchasing power for comparisons over time.

absolute well-being: The resources and opportunities a person has on their own: what they can afford or do.

access: The ability to obtain or use something.

average: A summary found by dividing a total by the number of people or items; it can hide differences.

benchmarks: Selected reference values used for comparisons.

bottom 50%: The half of a population with the lowest incomes in the comparison.

Btu: British thermal unit, a measure of energy.

capability: An ability or possibility: what people are able to do.

capacity: The amount a person or system can do or produce.

claim: A statement or explanation that needs support.

corroborate: Check whether another independent source supports a statement.

cotton gin: A machine that separates seeds from cotton fibers.

deprivation: A lack of resources needed for life or participation.

distribution: How a total is divided among people or groups.

enrolled: Officially registered to attend a school or program.

estimates: Values calculated from incomplete evidence or a model; they carry uncertainty.

evidence: Information used to support or question a claim.

functional literacy: The ability to use reading and writing effectively in everyday tasks.

GDP per capita: Gross domestic product divided by population: average economic output per person, not an individual's income.

Gilded Age: A name commonly used for the late nineteenth-century United States, associated with rapid growth and sharp inequalities.

hypothetical: Made up to explain an idea, rather than observed in historical evidence.

illiterate: Classified as unable to read or write under a source's definition.

implication: Something the evidence suggests may follow.

income share: The percentage of a group's total income received by a person or subgroup.

Industrial Revolution: The long shift toward machine production, new sources of power, and factory organization, beginning in eighteenth-century Britain.

industrialization: The process of expanding machine production, industrial energy use, and new forms of organizing work.

inequality: Unequal distribution of resources or opportunities.

infant mortality: Death before age one; the rate is usually deaths per 1,000 live births.

infrastructure: Shared systems and facilities, such as roads, water networks, and electricity lines.

institutions: Durable rules and organizations that shape people's choices, such as laws, schools, and systems of ownership.

labor: Human effort used to produce goods or services.

life expectancy: Estimated average years of life if a population experienced the death rates of a particular period.

literacy: The ability to read and write; different measures use different standards.

lumen-hours: A measure of light service: brightness in lumens multiplied by time in hours.

mechanism: The process connecting a possible cause to an outcome.

mechanisms: The processes connecting possible causes to outcomes.

median: The middle value when observations are ranked from lowest to highest.

metric: A specific measure used to describe or compare something.

metrics: Specific measures used to describe or compare things.

observations: Information recorded or measured, rather than assumed.

outcome: A result or consequence.

outcomes: Results or consequences.

output: The goods or services produced.

participation: Being able to take part and have a voice.

pattern: A recurring feature or direction in evidence.

percentage points: The difference between two percentages: 75% minus 51% is 24 percentage points.

perspective: A point of view shaped by a person's position and experience.

primary energy: Energy in its initial forms, before conversion into electricity or other usable forms.

production: The process of making goods or providing services.

productivity: Output for each unit of input, such as output per hour of labor.

public housing: Rental housing supported by public programs to help eligible households afford a home.

purchasing power: What money can buy. These adjustments help compare output across places with different prices.

quadrillion: One thousand trillion: 1,000,000,000,000,000.

qualify: Limit a claim to the conditions the evidence actually supports.

quintile: One fifth (20%) of a population ranked by a measure such as income.

ratio: A comparison made by dividing one amount by another.

reconstructions: Estimates of past conditions built from surviving evidence.

relative: Described in comparison with something else.

relative position: How a person's resources or opportunities compare with those of other people.

relative poverty: Income below a threshold tied to typical income in a society, often 50% of median household income adjusted for household size.

scheduled: Planned or required by a timetable.

security: Stability or protection against losing important resources and opportunities.

security and participation: Stability in important opportunities and the ability to take part in decisions.

steam engines: Machines that convert energy from heated water into mechanical motion.

thresholds: Cutoff points used to define a category.

top 10%: The tenth of a population with the highest incomes in the comparison.

transformation: A major change in how something works or is organized.

transmission: Sending information from one place to another.

trend: The broad direction of change over time.

unevenly: At different rates or to different degrees across people or places.

Finish the paper portion

Check all 39 response fields and eight vocabulary matches. Turn in this packet as your teacher directs. Then use the separate Boost interview-and-reflection document.